No.

Date

KARAKTERISTIK JFET ATAU AZUANA

A. TUJUAN

1. Menganalisis Idss dan Vgs off (Vp) dari JFET MPF102 pada rangkaian fixed bias dan memanfaatkan titik kerja pada daerah linier,
2. Menganalisis besarnya penguat tegangan (Av)

B. DASAR TEORI

Field Effect Transistor (JFET) merupakan uni polar transistor dengan sifat kerja tegangan sebagai kontrol device. Fet umumnya digunakan sebagai penguat, perbedaan antara FET dan BJT yaitu

1. Mempunyai resistansi input yang besar
2. Tahan terhadap radiasi, sedangkan BJT sangat sensitif (hfe).
3. Noisenya lebih rendah dari BJT, sehingga FET lebih banyak digunakan frekuensi tinggi

Ada 2 karakteristik FET yang penting, yaitu karakteristik drain yang merupakan kurva arus drain melawan tegangan Vgs dengan parameter Vgs sebagai variabel. Sedangkan yang satu yaitu kurva karakteristik transfer yang merupakan kurva arus drain yang melawan tegangan Vgs.

C. HASIL PENELITIAN

1. Rangkaian Pertama

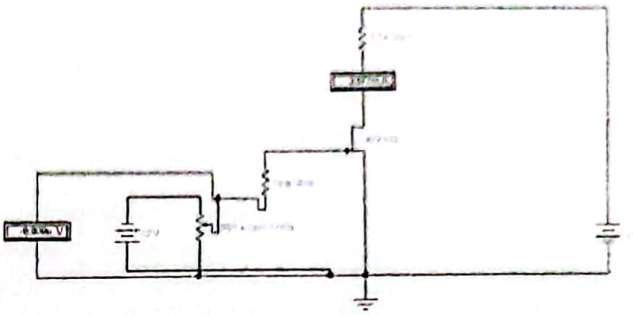
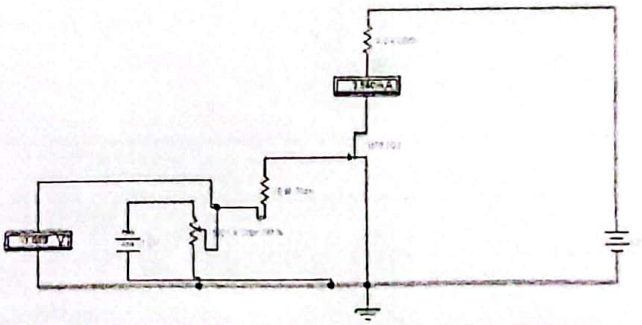
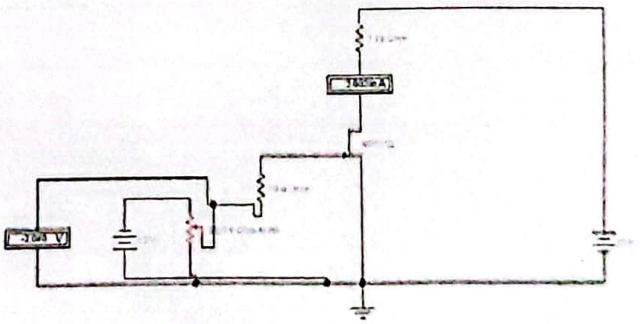
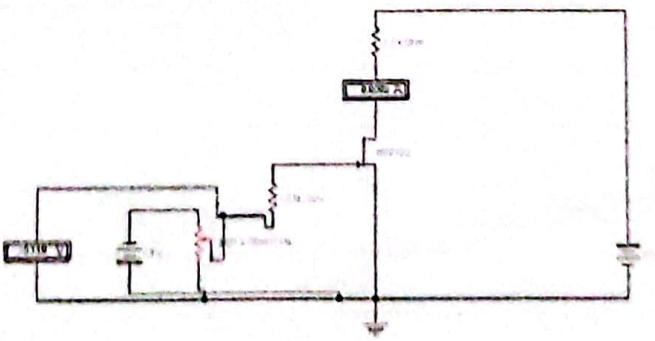
VGS (V)	ID (mA)
0	3,557
-1	3,540
-2	3
-3	0

2. Rangkaian Kedua

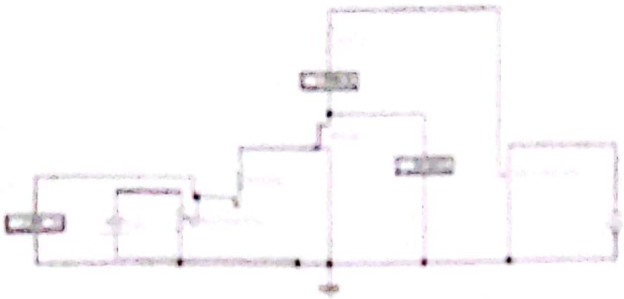
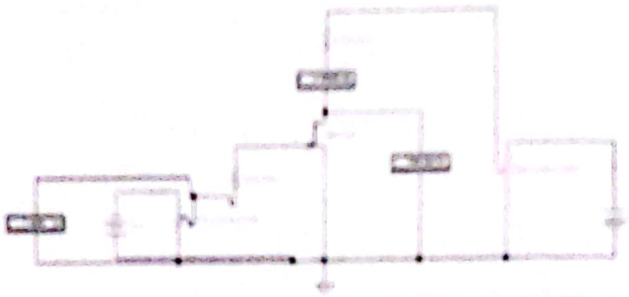
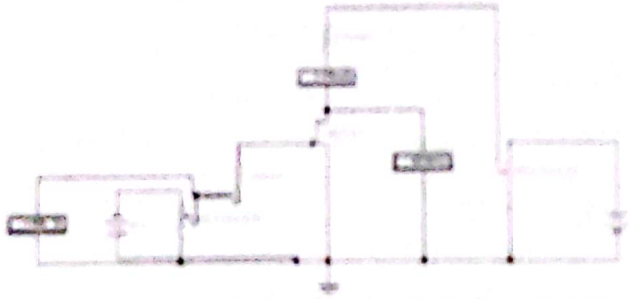
VGS\VDS	0,1	0,2	0,3	0,5	0,7	1	2	5	7	10	12
0	1,866 mA	3,577 mA									
-1	1,191 mA	2,3 mA	3,43 mA								
-2	596,8	1,132	1,571	2,287							

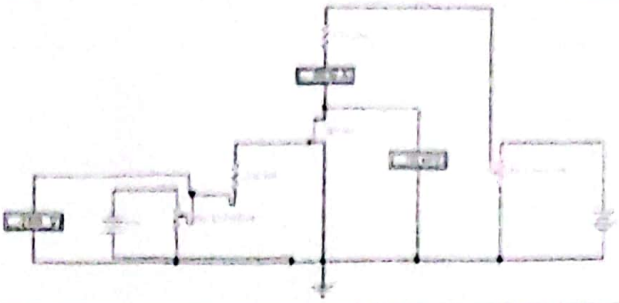
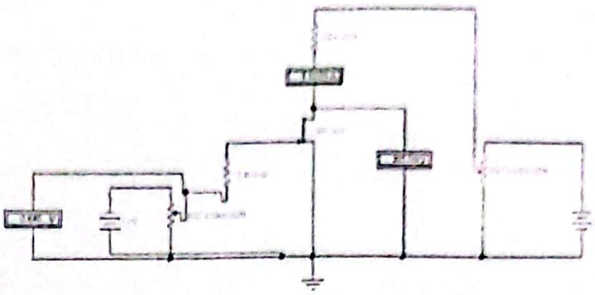
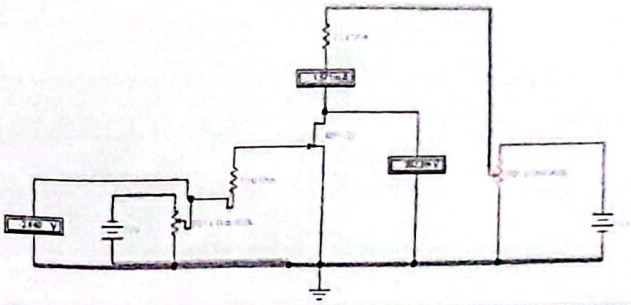
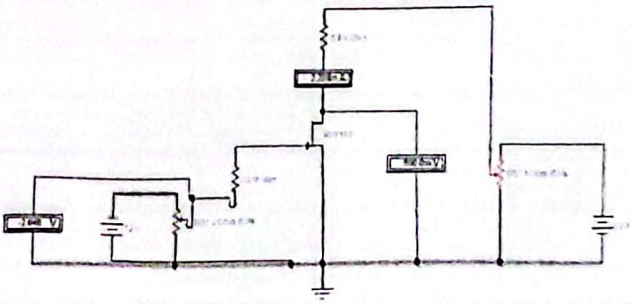
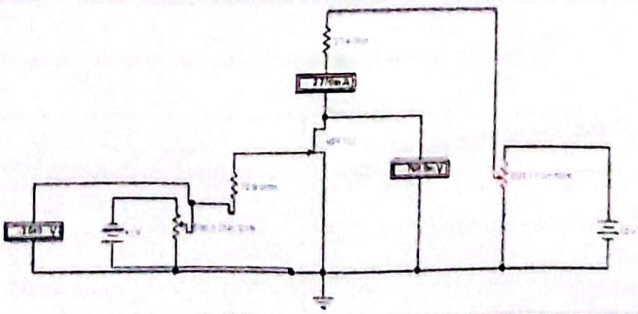
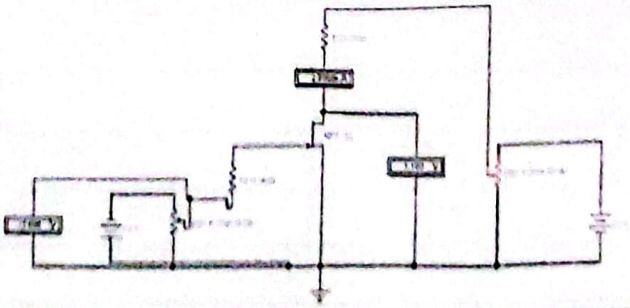
B. Dokumentasi Hasil Praktikum

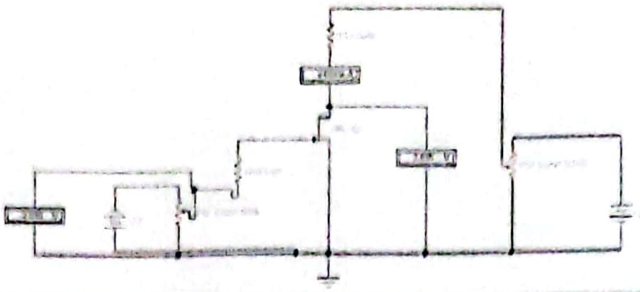
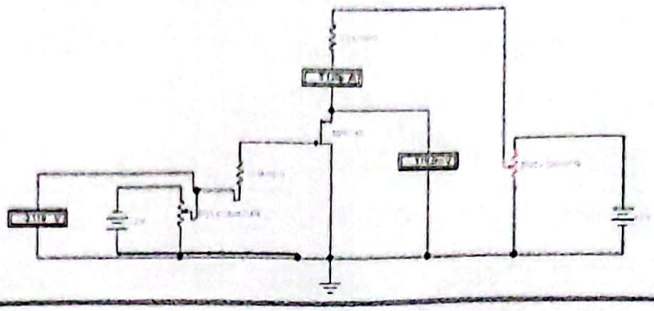
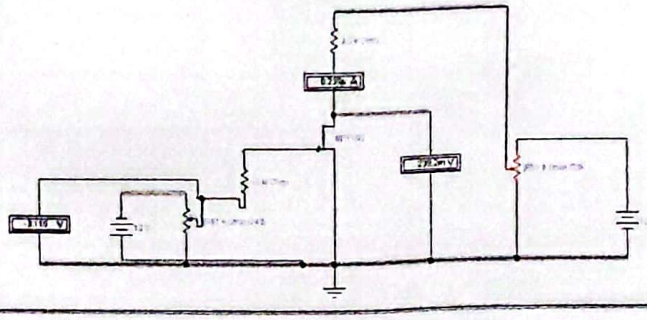
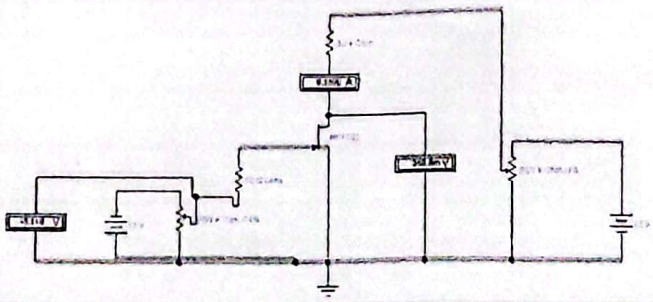
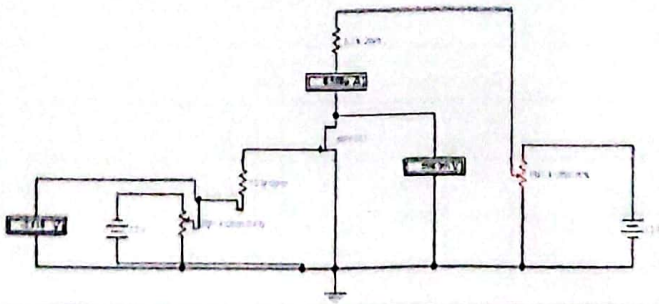
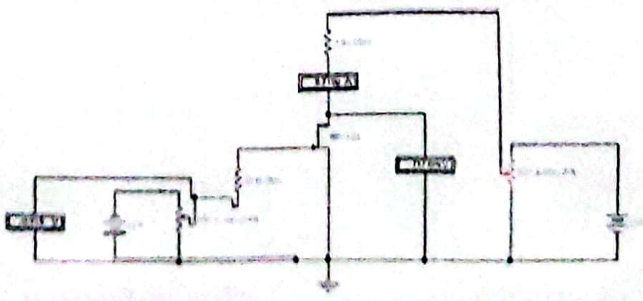
1. Rangkaian Pertama

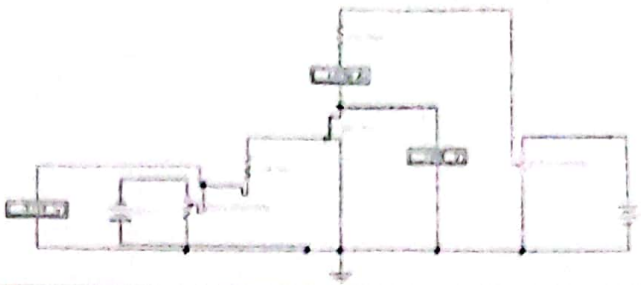
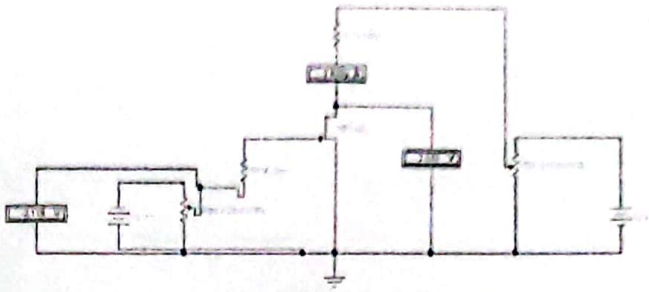
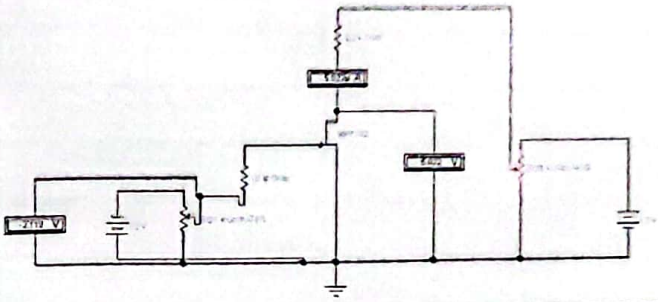
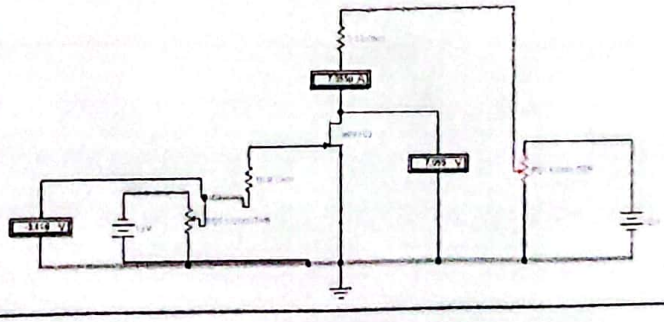
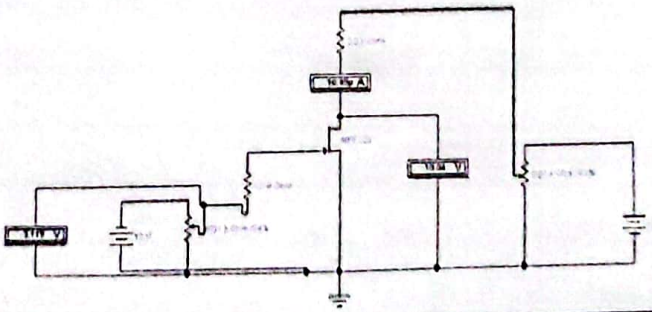
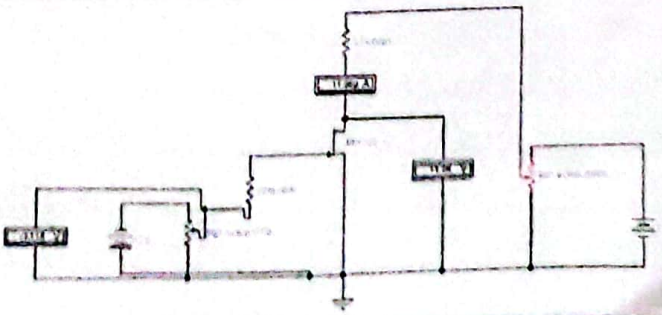
Keterangan	Dokumentasi
VGS = 0V ID = 3,557 mA	
VGS = -1V ID = 3,540 mA	
VGS = -2V ID = 3 mA	
VGS = -3V ID = 0 mA	

2. Rangkaian Kedua

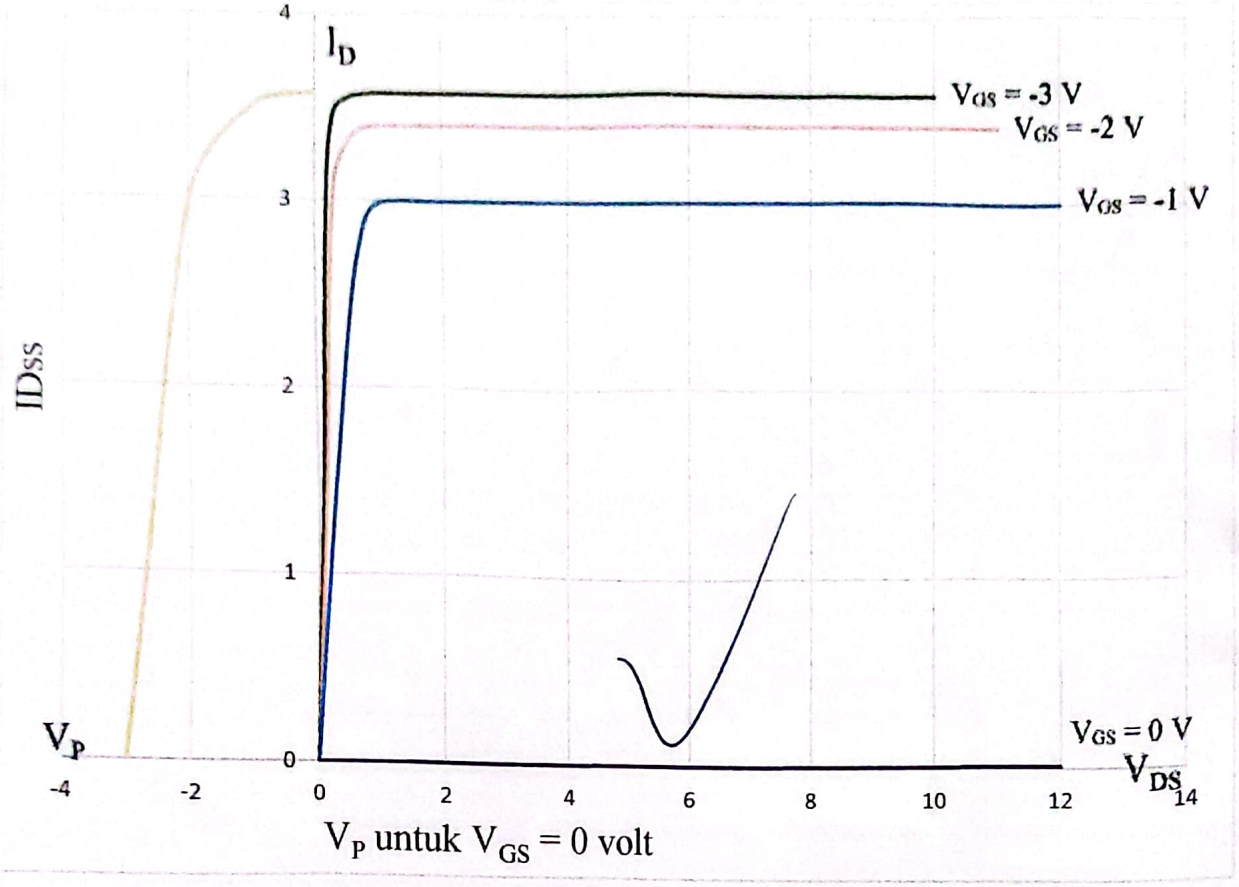
VGS = 0 V VDS = 0,1 V ID = 1,866 mA	
VGS = 0 V VDS = 0,2 V ID = 3,577 mA	
VGS = -1 V VDS = 0,1 V ID = 1,195 mA	
VGS = -1 V VDS = 0,2 V ID = 2,335 mA	
VGS = -1 V VDS = 0,3 V ID = 3,43 mA	

VGS = -2 V VDS = 0,1 V ID = 596,8 µA	
VGS = -2 V VDS = 0,2 V ID = 1,132 mA	
VGS = -2 V VDS = 0,3 V ID = 1,57 mA	
VGS = -2 V VDS = 0,5 V ID = 2,288 mA	
VGS = -2 V VDS = 0,7 V ID = 2,77 mA	
VGS = -2 V VDS = 1 V ID = 2,989 mA	

VGS = -2 V VDS = 2 V ID = 3 mA	
VGS = -3 V VDS = 0,1 V ID = 0,12 µA	
VGS = -3 V VDS = 0,2 V ID = 0,239 µA	
VGS = -3 V VDS = 0,3 V ID = 0,359 µA	
VGS = -3V VDS = 0,5 V ID = 0,598 µA	
VGS = -3 V VDS = 0,7 V ID = 0,718 µA	

VGS = -3 V VDS = 1 V ID = 1,076 µA	
VGS = -3 V VDS = 2 V ID = 2,033 µA	
VGS = -3 V VDS = 5 V ID = 5,022 µA	
VGS = -3 V VDS = 7 V ID = 7,055 µA	
VGS = -3 V VDS = 10 V ID = 10,05 µA	
VGS = -3 V VDS = 12 V ID = 11,96 µA	

C. Analisis Hasil Praktikum



V_P untuk $V_{GS} = 0$ volt

D. KESIMPULAN

Tegangan Gate & Source (V_{GS}) adalah tegangan pengontrol I_D & V_{DS} pada JFET. Kurva karakteristik input dari JFET berbentuk setengah parabola dengan sumbu x adalah V_{GS} dan sumbu y adalah I_D. Nilai arus maksimum disebut I_{DSS} = arus saturasi transistor, yaitu saat $V_{GS} = 0$. Ketika $I_D = 0$, maka $V_{GS} = V_P$.

Kurva karakteristik output akan mendekati konstan pada active region. Daerah aktif dan resistif dibatasi oleh garis $V_{DS} = |V_P|$.

7 mm ruled X 35 lines